Yonghao Shi

Ph.D. Student in Computer Science, Simon Fraser University

Human-Computer Interaction | Smart Environments | Computational Materials | Applied AI

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Research Profile

I study human-computer interaction systems that connect smart materials, room-scale sensing, fabrication, and machine learning. My recent work explores battery-free wood panels, RFID-augmented environments, vibrotactile rendering, smart furniture, and low-barrier prototyping platforms for interactive devices.

Education

2022-now	Ph.D. in Computer Science, Simon Fraser University, Burnaby, Canada. Thesis: Human-Computer Interaction: Smart Environment and Computational Material.
2017-2018	M.Sc. in Computer Science, University of Plymouth, Devon, United Kingdom. Thesis: Data Science and Business Analysis.
2016-2017	B.Sc. in Electrical and Electronic Engineering, University of Plymouth, Devon, United Kingdom.
2013-2016	B.Sc. in Electrical and Electronic Engineering, Beihang University, China.

Employment History

2020-2021	Project Manager, Institute of Information Engineering, Chinese Academy of Sciences.
2018-2020	Software Engineer, Institute of Information Engineering, Chinese Academy of Sciences.
2017.6-2017.9	Software Engineer, PwC Beijing.

Current Manuscripts Under Review

Lumberina: A Battery-Free, Wireless Smart Wood Panel with Concealed Dot-Matrix Display. Under review.

EcoVibWood: A Sustainable Vibration-Sensing Plywood Panel with Eco-Friendly Sensing Layers and a Self-Isolating Electrode Switching Mechanism. Under review.

Exploring Tangible Devices for Low-Effort, Participant-Controlled Data Input. Under review.

Peer-Reviewed Publications

November 2025	3Duino: A Low-Barrier Platform for Prototyping Interactive 3D-Printed Devices. ACM Symposium on Computational Fabrication. DOI: 10.1145/3745778.3766649.
May 2024	WooDowel: Electrode Isolation for Electromagnetic Shielding in Triboelectric Plywood Sensors. CHI Conference on Human Factors in Computing Systems. DOI: 10.1145/3613904.3642304.
May 2024	Tagnoo: Enabling Smart Room-Scale Environments with RFID-Augmented Plywood. CHI Conference on Human Factors in Computing Systems. DOI: 10.1145/3613904.3642356.
October 2023	Laser-Powered Vibrotactile Rendering. Adjunct Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology. DOI: 10.1145/3586182.3615795.

Skills

Coding

Java, C, Python, R, SQL, XML/XSL, LaTeX.

Databases

MySQL, Oracle, SQL Server, Redis, MongoDB, Neo4j.

Research

Academic research, training, consultation, HCI prototyping, data analysis.

Updated May 2026.